Schoonvelde et al. (2019):

Liberals lecture, conservatives communicate: Analyzing complexity and ideology in 381,609 political speeches

Replication

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TEXT ANALYSIS – P.S. 9599

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Introduction

- Does the well-documented finding that **liberal politicians use more complex language** than conservatives hold across countries, time periods and political contexts?
  - (previously only shown in US and UK)
- Analyzing **381 609 speeches** from:
  - 5 European parliaments (ParlSpeech)
  - 12 PMs (EUSpeech)
  - Party Congresses from Denmark and the Netherlands (Party Congresses)
- **They find that it does!**
  - Culturally liberal politicians use more complex languages than conservatives.
  - Economic left vs right doesn't have an effect.
  - Party Fixed Effects – When a party becomes more culturally conservative, their speech becomes less complex.
  - Speeches are becoming less complex over time.

PLOS ONE | ONE

RESEARCH ARTICLE

Liberals lecture, conservatives communicate: Analyzing complexity and ideology in 381,609 political speeches

Abstract

Introduction

Methods

A DESCRIPTION OF THE DATA AND EMPIRICAL STRATEGY USED IN THE ARTICLE

Datasets

- **ParlSpeech**
  - Speeches from members of 7 European parliaments
    - **British House of Commons** (N = 161,683, **1988–2015**)
    - **German Bundestag** (N = 66,061, **1991–2013**)
    - **Dutch Tweede Kamer** (N = 48,546, **1994-2015**)
    - **Spanish Congresso de Los Disputados** (N = 35,986, **1989-2015**)
    - **Swedish Riksdag** (N = 72,999, **1991-2015**)
- **EUSpeech**
  - All available speeches from elites in the main European institutions, the IMF, and speeches of prime ministers or presidents — or president in the case of France — of 10 EU member states for the period ranging from **early 2007 to late 2015**.
  - These countries are **Czech Republic, France, Germany, Greece, Netherlands, Italy, Spain, United Kingdom, Poland and Portugal**.
- **Party Congress Speeches**
  - The third dataset contains speeches at **party congresses in Denmark and the Netherlands, covering the time period 1945–2017**

Filtering Speeches

- **ParlSpeech**
  - Only speeches with ten sentences of at least five words (To avoid interruptions).
  - Exclude all chairmen/woman speeches as they serve to organize the debates.
- **EU Speech**
  - All English-language prime minister (PM) speeches.
  - Since we had only very few English speeches for Italian Prime Minister Prodi (3 speeches) and Portuguese Prime Minister Pedro Passos Coelho (6 speeches), we excluded them from this analysis.
- **Party Congresses**
  - We combined speeches from Christian Democratic Appeal (CDA) congresses and those of the three constituent parties ARP, CHU and KVP.

Complexity

- **Flesch-Kincaid Grade Level Scores:** $$score = 0.39 * (\frac{total\ words}{total\ sentences}) + 11.8 * \frac{total\ syllables}{total\ words} - 15.59$$

- Schoonvelde et al. mention: "Our approach does impede a comparison between countries, because **languages may systematically differ** in their complexity. It should be noted, however, that **our comparisons are within countries, not across countries**."

- Should also note that they **normalize the scores** for some parts of the analysis.

Ideology

- **Manifesto Project:** Systematically hand-coded issues in the election manifestos of parties.

- To calculate Liberal verses Conservative cultural dimension, they used this formula:

$$ideo = log(\sum conservative\ issues + 0.5) - log(\sum libeal\ issues + 0.5)$$

- To calculate Left vs Right economic dimension, they used this formula:

$$ideo = log(\sum left\ issues + 0.5) - log(\sum right\ issues + 0.5)$$

Cultural Issues

Conservative Issues:

- anti-EU
- anti-immigration
- pro-national way of life
- pro-traditional morality
- anti-multiculturalism
- pro-military
- anti-internationalism
- pro-Freedom and Human Right
- pro-political authority
- pro-law and order

Liberal Issues:

- pro-EU
- pro-immigration
- anti-national way of life
- anti-traditional morality
- pro-multiculturalism
- Antimilitary
- pro-internationalism
- anti-imperialism
- pro-peace
- pro-environment
- pro-culture
- support for under-privileged minority groups.

Economic Issues

Right issues:

- free-market economy
- incentives
- protectionism: negative
- economic growth: positive
- economic orthodoxy
- welfare state limitation
- labour groups: negative

Left Issues:

- market regulation
- economic planning
- corporatism
- protectionism: positive
- Keynesian demand management
- controlled economy
- nationalisation
- Marxist analysis
- welfare state expansion
- education expansion
- support for labour groups

Statistical Models

- **Standard OLS**
  - Regressing Flesch-Kincaid scores on:
    - Liberal / Conservative
    - Left / Right
    - Time
    - Currently in power
- Robustness: **Speaker Fixed Effects**
  - Same checks as the standard analysis

Results & Differences

- WHAT I WAS ABLE TO REPLICATE (TABLES, FIGURES, KEY QUANTITIES)
- ANY DIFFERENCES BETWEEN MY RESULTS AND THE AUTHORS' RESULTS

Descriptive Statistics

Author's Data

Table A.1: Party congresses DNK

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
Complexity	528	−0.046	0.771	−1.396	−0.513	0.315	6.435
Left-Right	528	−0.222	1.624	−3.925	−1.156	1.312	3.104
Progressive-Conservative	528	0.405	1.688	−2.938	−0.672	1.120	6.037
Year	528	44.670	15.520	0	33.8	57	69
In government	528	0.347	0.476	0	0	1	1

Table A.2: Party congresses NLD

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
Complexity	659	−0.057	0.697	−1.493	−0.583	0.418	2.486
Left-Right	659	−0.080	0.889	−2.011	−0.721	0.633	1.807
Progressive-Conservative	659	−0.582	0.905	−2.543	−1.140	−0.133	2.622
Year	659	39.420	20.514	1	22	58	72
In government	659	0.584	0.493	0	0	1	1

Table A.3: Parliament GER

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
Complexity	60,763	0.150	0.488	−2.030	−0.192	0.451	6.756
Left-Right	60,763	−0.326	1.185	−2.921	−1.235	0.740	1.485
Progressive-Conservative	60,763	−0.290	0.972	−2.912	−0.889	0.189	1.336
Year	60,763	12.342	6.714	0	7	19	22
In government	60,763	0.548	0.498	0	0	1	1

My Data

Table A.1: Party congresses DNK

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
Complexity	451	-0.045	1.081	-1.793	-0.654	0.379	11.297
Left-Right	451	0.926	1.714	-2.915	-0.393	1.950	4.263
Progressive-Conservative	451	0.302	1.348	-2.148	-0.565	0.837	4.710
Year	451	45.550	15.494	0.000	35.000	58.000	69.000
In government	451	0.118	0.322	0.000	0.000	0.000	1.000

n = -71

Table A.2: Party congresses NLD

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
Complexity	540	0.038	0.926	-1.913	-0.621	0.653	3.269
Left-Right	540	0.872	0.866	-1.784	0.246	1.455	3.120
Progressive-Conservative	540	-0.515	0.641	-1.867	-0.978	-0.110	2.446
Year	540	37.350	21.076	0.000	19.750	57.000	71.000
In government	540	0.219	0.414	0.000	0.000	0.000	1.000

n = -119

Table A.3: Parliament GER

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
Complexity	61,512	0.120	0.488	-1.866	-0.220	0.423	4.651
Left-Right	61,512	1.056	1.123	-0.629	0.152	2.034	3.391
Progressive-Conservative	61,512	-0.313	0.822	-2.435	-0.786	0.117	1.194
Year	61,512	12.104	6.711	0.000	6.000	18.000	22.000
In government	61,512	0.515	0.500	0.000	0.000	1.000	1.000

n = +749

Descriptive Statistics Cont.

Author's Data

Table A.4: Parliament ESP

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
Complexity	35,154	2.502	1.051	−1.795	1.786	3.143	9.028
Left-Right	35,154	−0.213	0.790	−2.239	−0.859	0.495	1.234
Progressive-Conservative	35,154	−0.853	0.662	−2.576	−1.397	−0.300	0.206
Year	35,154	13.212	7.264	0	7	20	26
In government	35,154	0.371	0.483	0	0	1	1

Table A.5: Parliament GBR

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
Complexity	161,439	−0.178	0.465	−2.716	−0.461	0.109	8.640
Left-Right	161,439	0.146	0.705	−0.841	−0.448	0.684	1.666
Progressive-Conservative	161,439	0.527	0.745	−1.124	−0.028	0.839	1.844
Year	161,439	14.014	7.718	0	7	21	27
In government	161,439	0.545	0.498	0	0	1	1

Table A.6: Parliament SWE

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
Complexity	72,955	−0.728	0.414	−3.088	−1.006	−0.472	2.093
Left-Right	72,955	0.152	1.364	−3.260	−0.831	0.891	3.613
Progressive-Conservative	72,955	−0.501	0.998	−5.112	−0.995	0.254	2.056
Year	72,955	13.051	7.294	0	7	19	25
In government	72,955	0.391	0.488	0	0	1	1

My Data

Table A.4: Parliament ESP

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
Complexity	34,622	2.077	1.004	-1.062	1.377	2.701	8.214
Left-Right	34,622	1.099	0.789	-0.400	0.515	1.659	2.898
Progressive-Conservative	34,622	-0.699	0.551	-2.637	-1.182	-0.281	0.162
Year	34,622	13.660	7.313	0.000	8.000	20.000	26.000
In government	34,622	0.346	0.476	0.000	0.000	1.000	1.000

n = -532

Table A.5: Parliament GBR

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
Complexity	124,363	-0.197	0.737	-2.369	-0.640	0.163	78.973
Left-Right	124,363	0.805	0.725	-1.121	0.214	1.464	1.751
Progressive-Conservative	124,363	0.367	0.612	-1.547	0.000	0.780	2.998
Year	124,363	14.226	7.655	0.000	8.000	21.000	27.000
In government	124,363	0.498	0.500	0.000	0.000	1.000	1.000

n = -37 076

Table A.6: Parliament SWE

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
Complexity	65,535	-0.573	0.509	-2.452	-0.930	-0.246	2.146
Left-Right	65,535	0.428	1.371	-2.160	-0.388	1.387	3.660
Progressive-Conservative	65,535	-0.545	0.909	-4.117	-1.103	0.197	1.587
Year	65,535	13.270	7.258	0.000	7.000	19.000	25.000
In government	65,535	0.382	0.486	0.000	0.000	1.000	1.000

n = -7 420

Descriptive Statistics Cont.

Author's Data

Table A.7: Parliament NLD

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
Complexity	48,281	−0.383	0.378	−2.109	−0.629	−0.175	4.625
Left-Right	48,281	−0.140	0.929	−3.100	−0.672	0.403	1.469
Progressive-Conservative	48,281	0.006	0.991	−2.613	−0.739	0.593	2.622
Year	48,281	11.664	6.098	0	6	17	21
In government	48,281	0.551	0.497	0	0	1	1

Table A.8: Heads of Government

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
Complexity	1,830	0.089	0.756	−2.210	−0.446	0.547	5.865
Left-Right	1,830	0.010	0.760	−2.356	−0.921	0.684	1.102
Progressive-Conservative	1,830	−0.079	1.005	−3.628	−0.487	0.828	3.110

My Data

Table A.7: Parliament NLD

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
Complexity	44,258	-0.389	0.544	-2.186	-0.742	-0.091	6.266
Left-Right	44,258	0.963	0.789	-0.350	0.364	1.357	3.382
Progressive-Conservative	44,258	0.103	0.818	-2.043	-0.399	0.594	2.446
Year	44,258	11.692	6.100	0.000	6.000	17.000	21.000
In government	44,258	0.486	0.500	0.000	0.000	1.000	1.000

n = -4023

Table A.8: Heads of Government

Statistic	N	Mean	St. Dev.	Min	Pctl(25)	Pctl(75)	Max
Complexity	1,442	0.000	1.000	-2.886	-0.718	0.650	4.341
Left-Right	1,442	0.919	0.705	-0.473	0.214	1.659	2.584
Progressive-Conservative	1,442	-0.054	0.836	-2.059	-0.387	0.780	2.312
Year	1,442	4.394	2.469	0.000	2.000	7.000	8.000
In government	1,442	0.917	0.276	0.000	1.000	1.000	1.000

n = -388

Figure 1: Descriptive Information on Linguistic Complexity

Author's Data

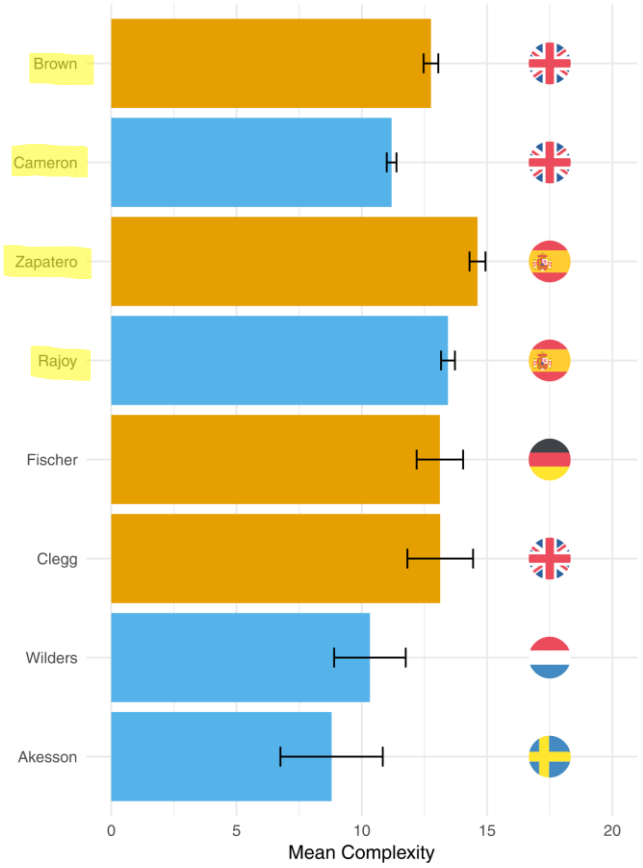


Fig 1. Descriptive information on linguistic complexity. The bars in this figure denote mean complexity scores, with 95% confidence intervals, of selected speakers.

https://doi.org/10.1371/journal.pone.0208450.g001

My Data

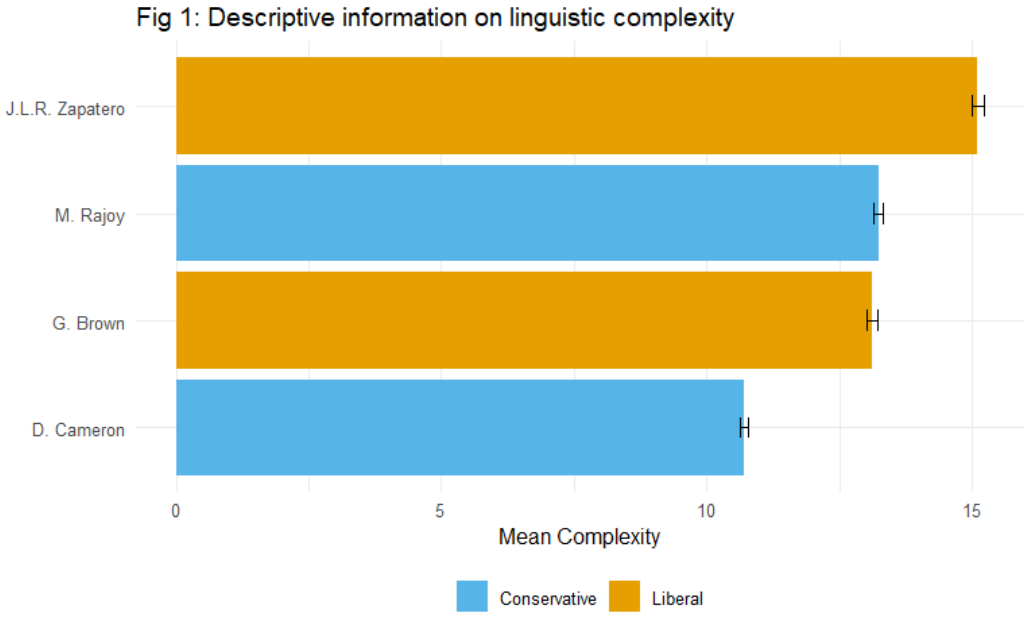


Figure 2: OLS Regression on Ideology

Author's Data

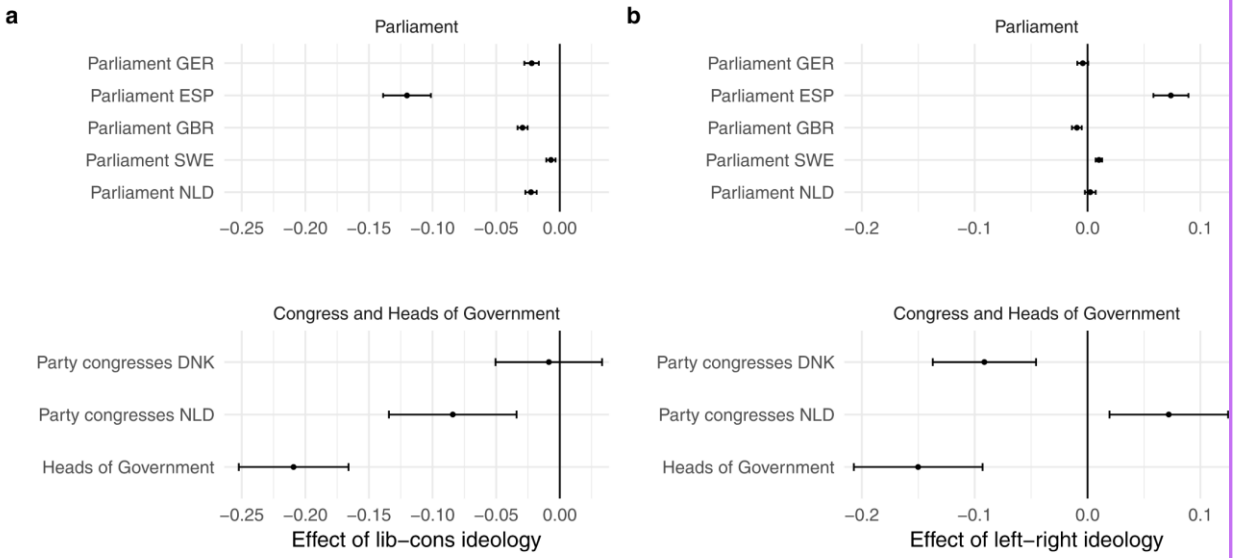


Fig 2. OLS regression of complexity on ideology. Plot *a* reports standardized regression coefficients for liberal-conservative ideology in eight OLS regression models (one for each dataset in the corpus). Plot *b* reports standardized regression coefficients for left-right ideology. The lines represent the 95% confidence intervals of the coefficients.

https://doi.org/10.1371/journal.pone.0208450.g002

My Data

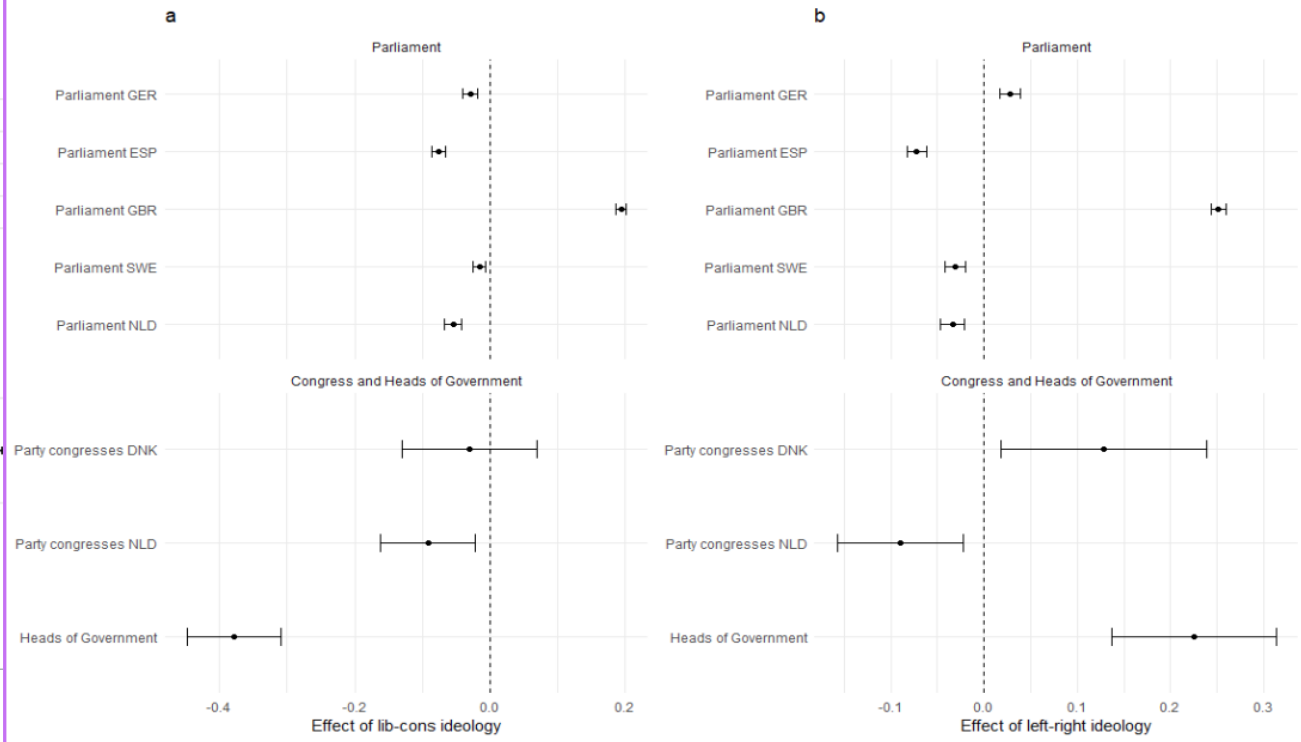


Figure 3: OLS regression on complexity on time

Author's Data

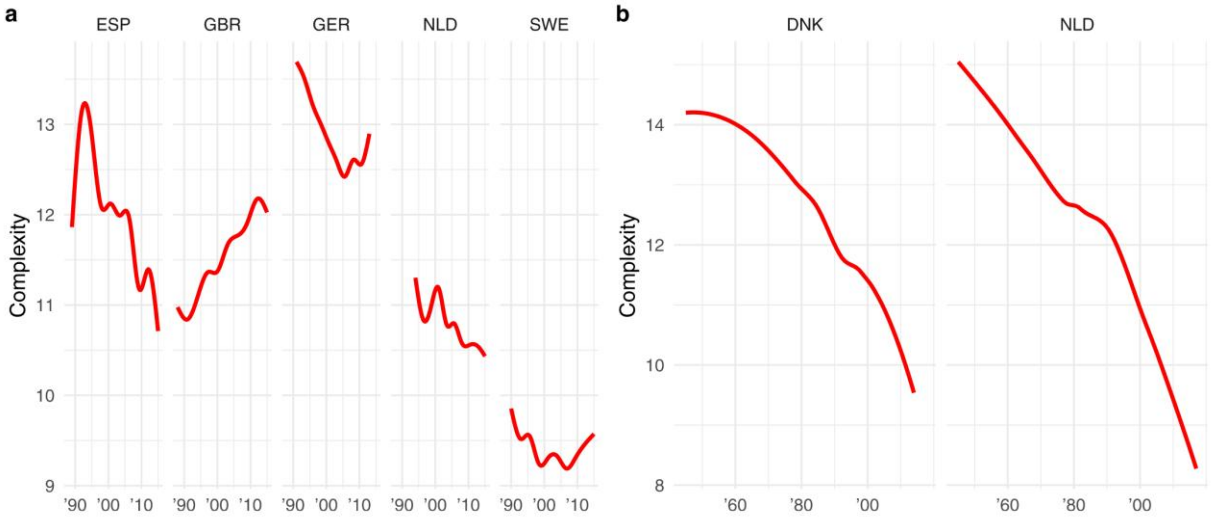


Fig 3. OLS regression of complexity on time. Plots *a* an *b* display loess regression lines of average speech complexity over time in the parliamentary speeches and congress speeches respectively. These are local estimates of the effect of year on complexity.

https://doi.org/10.1371/journal.pone.0208450.g003

My Data

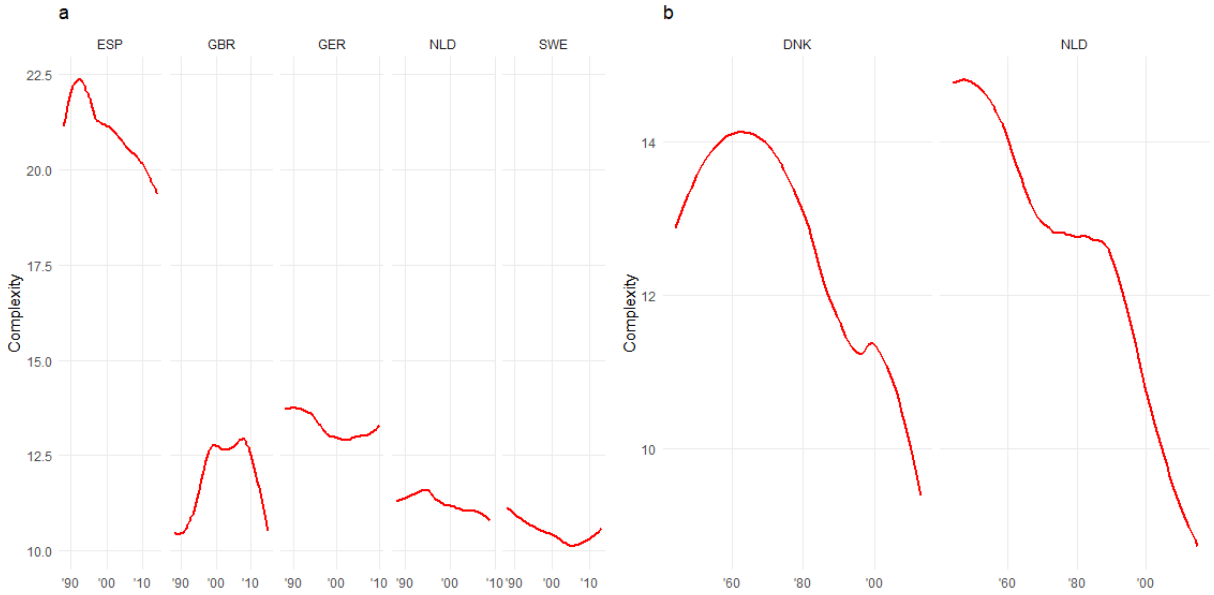


Figure 4: OLS regression on complexity on time

Author's Data

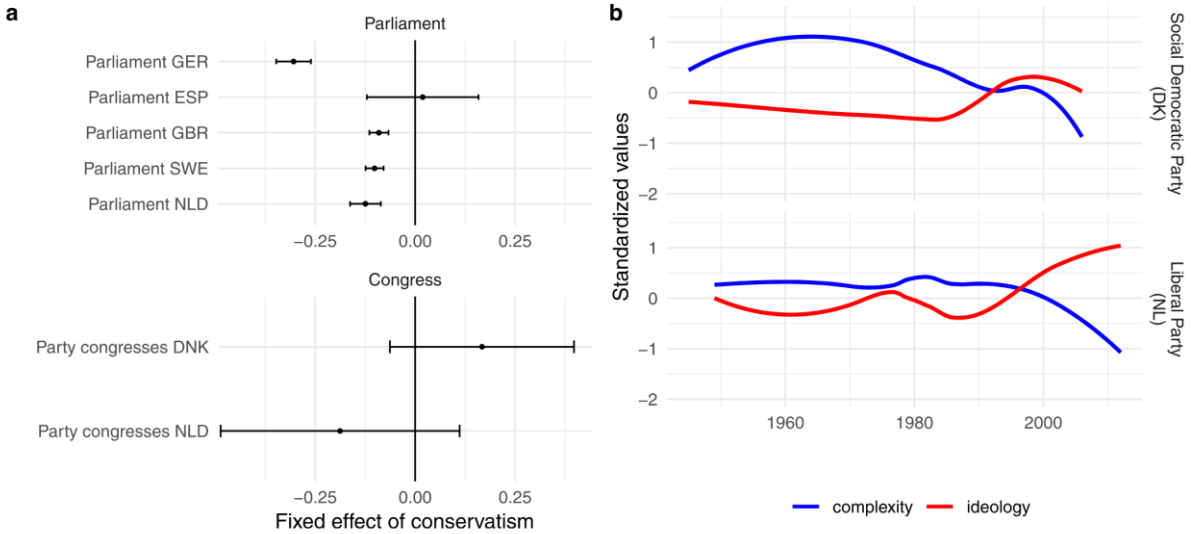
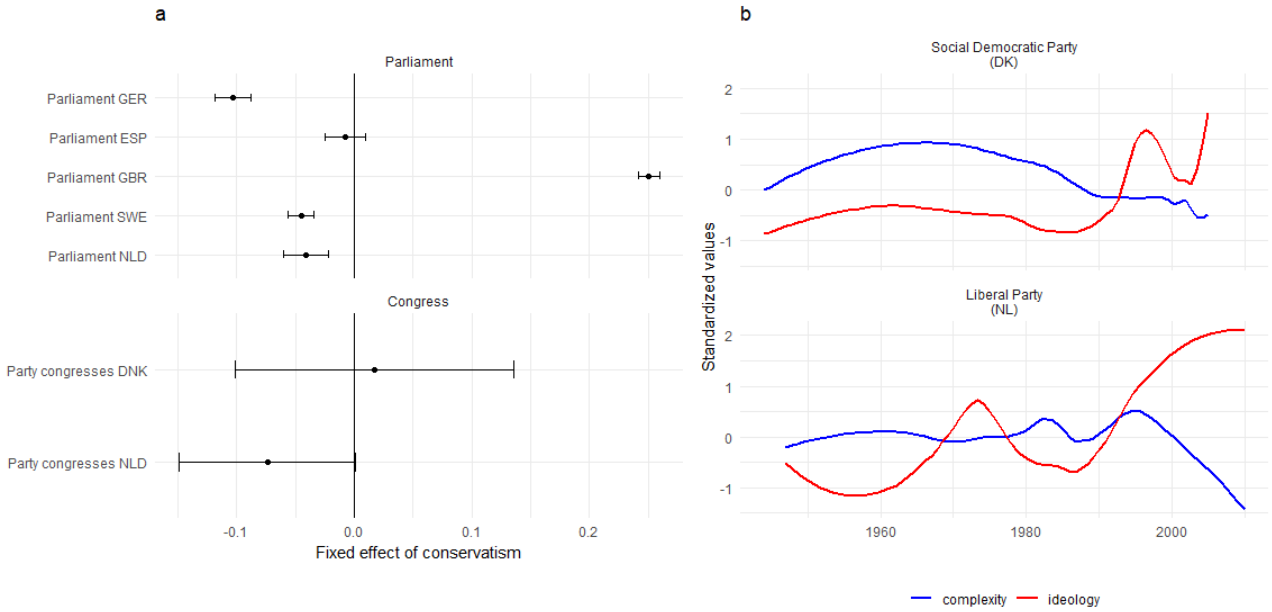


Fig 4. Regression of complexity on ideology with fixed effects for party. Plot *a* reports regression coefficients for liberal conservative ideology in 7 regression models with fixed effects for party (one for each dataset in the corpus with the exception of the prime minister speeches). Plot *b* displays loess regression lines of average speech complexity over time for the Danish social democratic party and the Dutch liberal party, as well as their ideological position.

https://doi.org/10.1371/journal.pone.0208450.g004

My Data



OLS Results (Party Fixed Effects)

Correct Significance Level + Right Direction
Right Direction + Wrong Significance Level
Wrong Direction (Possibly even significantly so)

Table C.1: OLS regression results

	DNK Congress	NLD Congress	GER Parl	ESP Parl
Left-Right	−0.091***	0.072***	−0.004*	0.074***
	(0.023)	(0.027)	(0.002)	(0.008)
Progressive-Conservative	−0.009	−0.084***	−0.022***	−0.120***
	(0.021)	(0.026)	(0.003)	(0.010)
Year	−0.021***	−0.022***	−0.009***	−0.020***
	(0.002)	(0.001)	(0.0003)	(0.001)
In government	0.159**	0.075*	0.039***	−0.020*
	(0.066)	(0.044)	(0.004)	(0.012)
Constant	0.836***	0.736***	0.235***	2.682***
	(0.113)	(0.061)	(0.005)	(0.015)
Observations	528	659	60,763	35,154
R^2	0.181	0.481	0.022	0.026
Adjusted R^2	0.175	0.478	0.022	0.026
Residual Std. Error	0.700 (df = 523)	0.504 (df = 654)	0.483 (df = 60758)	1.037 (df = 35149)
F Statistic	28.868*** (df = 4; 523)	151.337*** (df = 4; 654)	348.687*** (df = 4; 60758)	235.073*** (df = 4; 35149)
Note:				*p<0.1; **p<0.05; ***p<0.01

	GBR Parl	SWE Parl	NLD Parl	PM
Left-Right	−0.010***	0.010***	0.002	−0.150***
	(0.002)	(0.001)	(0.002)	(0.029)
Progressive-Conservative	−0.029***	−0.007***	−0.023***	−0.209***
	(0.002)	(0.002)	(0.002)	(0.022)
Year	0.014***	−0.001**	−0.007***	
	(0.0001)	(0.0003)	(0.0003)	
In government	−0.058***	0.071***	0.038***	
	(0.002)	(0.003)	(0.004)	
Constant	−0.328***	−0.753***	−0.326***	0.074***
	(0.003)	(0.005)	(0.005)	(0.016)
Observations	161,439	72,955	48,281	1,830
R^2	0.065	0.009	0.020	0.157
Adjusted R^2	0.065	0.009	0.020	0.156
Residual Std. Error	0.450 (df = 161434)	0.412 (df = 72950)	0.375 (df = 48276)	0.695 (df = 1827)
F Statistic	2,809.898*** (df = 4; 161434)	164.229*** (df = 4; 72950)	246.653*** (df = 4; 48276)	169.938*** (df = 2; 1827)
Note:				*p<0.1; **p<0.05; ***p<0.01

Table C.1: OLS regression results

	Dependent variable:			
	flesch_kincaid			
	DNK Congress	NLD Congress	GER Parl	ESP Parl
	(1)	(2)	(3)	(4)
Left-Right	0.081**	-0.096***	0.012***	-0.092***
	(0.035)	(0.037)	(0.003)	(0.007)
Progressive-Conservative	-0.024	-0.133***	-0.017***	-0.139***
	(0.041)	(0.051)	(0.003)	(0.010)
Year	-0.031***	-0.029***	-0.007***	-0.028***
	(0.004)	(0.002)	(0.0003)	(0.001)
In government	0.085	0.088	0.034***	-0.052***
	(0.150)	(0.085)	(0.004)	(0.011)
Constant	1.267***	1.106***	0.174***	2.478***
	(0.157)	(0.072)	(0.006)	(0.015)
Observations	451	540	61,512	34,622
R^2	0.162	0.451	0.016	0.054
Adjusted R^2	0.154	0.447	0.016	0.054
Residual Std. Error	0.994 (df = 446)	0.689 (df = 535)	0.484 (df = 61507)	0.976 (df = 34617)
F Statistic	21.520*** (df = 4; 446)	109.809*** (df = 4; 535)	246.720*** (df = 4; 61507)	497.892*** (df = 4; 34617)
Note:				*p<0.1; **p<0.05;** p<0.01

Table C.1 continued

	Dependent variable:			
	flesch_kincaid			
	GBR Parl	SWE Parl	NLD Parl	PM
	(1)	(2)	(3)	(4)
Left-Right	0.255***	-0.011***	-0.023***	0.252***
	(0.004)	(0.002)	(0.004)	(0.048)
Progressive-Conservative	0.233***	-0.009***	-0.036***	-0.442***
	(0.004)	(0.003)	(0.004)	(0.041)
Year	0.006***	-0.006***	-0.008***	
	(0.0003)	(0.0004)	(0.0004)	
In government	-0.021***	0.087***	-0.032***	
	(0.004)	(0.004)	(0.006)	
Constant	-0.563***	-0.532***	-0.258***	-0.255***
	(0.006)	(0.006)	(0.008)	(0.048)
Observations	124,363	65,535	44,258	1,442
R^2	0.054	0.018	0.011	0.267
Adjusted R^2	0.054	0.018	0.011	0.266
Residual Std. Error	0.717 (df = 124358)	0.504 (df = 65530)	0.541 (df = 44253)	0.857 (df = 1439)
F Statistic	1,765.100*** (df = 4; 124358)	306.168*** (df = 4; 65530)	122.660*** (df = 4; 44253)	261.871*** (df = 2; 1439)

OLS Results (Speaker Fixed Effects)

Lots are correctly significant and in right direction, but still wildly off the author's estimate

	DNK Congress	NLD Congress	GER Parl	ESP Parl
Left-Right	0.364**	0.141	0.019	−0.133**
	(0.154)	(0.192)	(0.020)	(0.061)
Progressive-Conservative	0.167	−0.188	−0.304***	0.019
	(0.117)	(0.152)	(0.022)	(0.071)
Year	−0.067***	−0.087***	−0.027***	−0.082***
	(0.010)	(0.005)	(0.001)	(0.003)
In government	1.210***	0.234	0.088***	0.852***
	(0.274)	(0.187)	(0.017)	(0.052)
Constant	15.104***	14.672***	13.280***	24.207***
	(0.716)	(0.304)	(0.025)	(0.210)
Party Fixed Effects?	*Yes*	*Yes*	*Yes*	*Yes*
Observations	528	659	60,763	35,154
R^2	0.239	0.500	0.027	0.064
Adjusted R^2	0.227	0.490	0.027	0.064
Residual Std. Error	2.652 (df = 519)	1.948 (df = 645)	1.885 (df = 60754)	3.980 (df = 35141)
F Statistic	20.367*** (df = 8; 519)	49.557*** (df = 13; 645)	211.974*** (df = 8; 60754)	200.657*** (df = 12; 35141)
Note:				*p<0.1; **p<0.05; ***p<0.01

	GBR Parl	SWE Parl	NLD Parl
Left-Right	−0.072***	0.037***	0.0003
	(0.014)	(0.009)	(0.016)
Progressive-Conservative	−0.091***	−0.102***	−0.124***
	(0.012)	(0.011)	(0.020)
Year	0.054***	−0.005***	−0.028***
	(0.001)	(0.001)	(0.001)
In government	−0.198***	0.302***	0.038**
	(0.010)	(0.013)	(0.018)
Constant	10.866***	9.178***	11.113***
	(0.014)	(0.030)	(0.028)
Party Fixed Effects?	*Yes*	*Yes*	*Yes*
Observations	161,439	72,955	48,281
R^2	0.068	0.022	0.036
Adjusted R^2	0.068	0.022	0.035
Residual Std. Error	1.759 (df = 161430)	1.602 (df = 72942)	1.455 (df = 48264)
F Statistic	1,464.877*** (df = 8; 161430)	136.353*** (df = 12; 72942)	111.615*** (df = 16; 48264)
Note:			*p<0.1; **p<0.05; ***p<0.01

Table D.1: Speaker fixed effects regression results

	Dependent variable:			
	flesch_kincaid			
	DNK Congress	NLD Congress	GER Parl	ESP Parl
	(1)	(2)	(3)	(4)
Left-Right	0.049	0.080	-0.020***	0.023
	(0.056)	(0.066)	(0.005)	(0.016)
Progressive-Conservative	0.012	0.036	-0.055***	-0.064***
	(0.052)	(0.067)	(0.005)	(0.018)
Year	-0.027***	-0.023***	-0.006***	-0.037***
	(0.008)	(0.004)	(0.0005)	(0.001)
In government	0.205	-0.056	0.014***	0.230***
	(0.219)	(0.102)	(0.005)	(0.016)
Constant	-0.536	1.484***	-0.078	2.787***
	(0.943)	(0.485)	(0.066)	(0.486)
Speaker Fixed Effects?	Yes	Yes	Yes	Yes
Observations	451	540	61,512	34,622
R^2	0.619	0.850	0.304	0.326
Adjusted R^2	0.321	0.740	0.283	0.301
Residual Std. Error	0.891 (df = 253)	0.472 (df = 312)	0.413 (df = 59700)	0.839 (df = 33363)
F Statistic	2.082*** (df = 197; 253)	7.770*** (df = 227; 312)	14.410*** (df = 1811; 59700)	12.842*** (df = 1258; 33363)
Note:				*p<0.1; **p<0.05;* p<0.01

Table D.1 continued

	Dependent variable:		
	flesch_kincaid		
	GBR Parl	SWE Parl	NLD Parl
	(1)	(2)	(3)
Left-Right	0.306***	-0.009***	0.013*
	(0.006)	(0.003)	(0.008)
Progressive-Conservative	0.260***	-0.009**	-0.002
	(0.006)	(0.004)	(0.007)
Year	0.001*	-0.014***	-0.019***
	(0.0004)	(0.001)	(0.001)
In government	-0.008	0.065***	-0.041***
	(0.005)	(0.006)	(0.009)
Constant	-0.636***	-0.399***	-0.588***
	(0.102)	(0.076)	(0.112)
Speaker Fixed Effects?	Yes	Yes	Yes
Observations	124,363	65,535	44,258
R^2	0.176	0.241	0.142
Adjusted R^2	0.166	0.218	0.128
Residual Std. Error	0.673 (df = 122852)	0.450 (df = 63560)	0.508 (df = 43561)
F Statistic	17.407*** (df = 1510; 122852)	10.242*** (df = 1974; 63560)	10.375*** (df = 696; 43561)
Note:			*p<0.1; **p<0.05;* p<0.01

Main Findings – Comparison

- **Liberal / Conservative – Cultural Dimension**
  - Author:
    - In 7 out of the eight corpora, conservative parties are associated with significant lower speech complexity. (Denmark being the exception).
  - Replication:
    - I got exactly this, except for my Great Britian Parliament dataset – likely due to loss of observations

- **Economic Left – Right**
  - Author
    - No systemic pattern
  - Replication:
    - No pattern either. Though the lack of pattern looks differently between the two.

- **Party Fixed Effects**
  - Author:
    - When accounting for party's fixed effects, the authors find that parties that shift to a more culturally conservative direction over time have increasingly simple speeches.
  - Replication:
    - My results replicated this.

- **Time**
  - Author
    - Find broad downward trend in speech complexity over time
  - Replication:
    - My results replicate this.

Replication Autopsy

WHAT WORKED, WHAT DID NOT, AND WHERE THINGS GOT STUCK

Obtaining Data

- **Author Data**
  - Author **doesn't include raw data**, so we cannot be sure of how they cleaned it. For this reason, I had to find the original data in references and start from scratch.
  - Though I was able to get the original data, this doesn't mean that the author's made it easy for me. Because they don't include the cleaning data, and they don't go through every little choice they made, I had to take some **assumptions** regarding the cleaning.
- **Original Data**
  - Downloaded from Harvard Dataverse:
    - ParlSpeech: https://doi.org/10.7910/DVN/E4RSP9
    - EUSpeech: https://doi.org/10.7910/DVN/XPCVEI (link in paper)
      - https://doi.org/10.7910/DVN/GKABNU (correct link)
    - Party Congresses: https://doi.org/10.7910/DVN/SFZI8O
  - Manifesto Project (Needed to Register): https://manifesto-project.wzb.eu/

Pre-Cleaning

- The first thing I did was open up the data and check it to make sure it all looks good and to understand it's structure. **Then I removed a lot of the variables that we didn't need for analysis**, keeping things cleaner.

- A useful thing I also did was to allow to **sample the data** instead of using the full amount. The processing started taking a really long time (especially with ParlSpeech's over 3 million) speeches at before filtering).
  - I used this for testing before running the full dataset for the results. (Doesn't mean I only did it once…)

Cleaning - ParlSpeech

- **Goal:** Remove all chairperson speeches, and interruptions (short speeches)

- This was relatively simple, chair person speeches had their own dummy variable, and I was able to use some Regex code to find sentences. It was simpler than I expected it to be.
  - **Still took a long time with such a large data set, about 7 minutes.**

- I also started matching the party names with the ones that are coming for the Manifesto Project ideology merge.

Cleaning – EUSpeech

- **Goal:** All English language speeches, no translations. Remove Italian PM Prodi and Portuguese PM Coelho.
  - **Discovered Goals:** Add party information. Wrong version (only pre-cleaned in 2016 version). Language (all texts were translated).

- Started terribly because I used wrong dataset. This is because the wrong was linked in the references (?) and it didn't have a non-cleaned version. Found updated one without stemming or stop words removed.
  - Original one I was using had translated texts as well. Making English only filtering do basically nothing.

- One "easy" part was creating database of speaker information and their parties for the manifesto project, which wasn't included in either version of the dataset.

Cleaning – Party Congresses

- **Goal:** Focus only on mentioned parties. Merge ARP, CHU and KVP into the Christian Democratic Appeal.

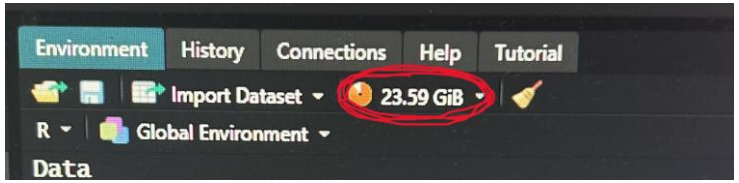
- This went rather smoothly. I just had to filter the unwanted parties, as well as merging those specific parties into the CDA.
  - Or at least I thought… though those parties did become the CDA, we still needed to calculate their ideological scores before the CDA existed.
    - This matters because the ideological scores are updated by election, so you can't just take one score and apply it over time.
  - Added some logic to figure this out. It uses the CDA when supposed to and uses others before it existed when calculating ideology.

Cleaning – Manifesto Project

- **Goal:** Calculate ideological scores of parties based on formula and issue typology. Add ideology to other datasets.

- Went mostly well. Had to figure out matching party names with the ones in the dataset but overall, it went well.
  - Had to use lots of tables to diagnose issues.

Cleaning – Complexity



- **Goal:** Calculate and add Flesch Kincaid grade scores.

- Used Quanteda function which made the actual scoring relatively easy.
- **Defeat? Or not…**
  - This is when I realized how big of an issue the original EUSpeech dataset was. It had removed all punctuation, so it was impossible to get accurate scores. So, I went back, found the right data that WAS available at the time, and reapplied all of the cleaning. It worked. I felt super defeated, thinking I was going to have to use the authors pre-cleaned data.
- Though it worked well on the sample (testing), when trying to apply it to the full dataset, it crashed my computer. So instead, I had to chunk it into 3 separate pieces which I than ran in succession. This worked fine.
- I had to standardize the values as they did in the paper, seeing in the descriptive statistics that my values were much higher.

Cleaning – Governing Party

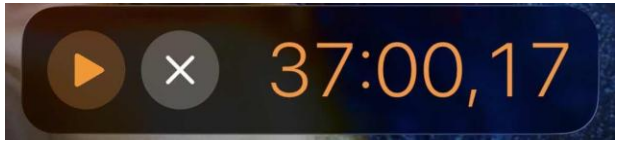
- Goal: Add dummy variable to indicate if the speaker was a member of the governing party.

- This worked relatively well and straight forward. I just had to make a table of which party was in power from which years, and check it against the speakers party.

Final Cleaning

- Now that I had added all of the important logic and information, I wanted to clean and save the data into a final dataset, as well as the country specific data.

- This was relatively straight forward, I was now ready to apply it to the full data and save my full data, final replication dataset…

- 

Analysis

- Once I had the data, the analysis was simple. I just had to replicate the tables and figures, run the OLSs and interpret the results. This was so much easier than the cleaning.

Autopsy Results

- Overall, I think the replication was mostly a success.
- Considering the authors didn't include their cleaning code, I had to make a lot of assumptions about some of the choices they made. Clearly I made some of them wrong, as I have different observation numbers and results…
- But not too different, most of the main takeaways are the same.

- Overall, I think the replication worked out well. I learned a lot about how long and hard it can be to clean data, and merging data from different datasets. I also learned that you should include all of the code and data when uploading your files for replication.
  - My GitHub repository will be much cleaner and complete

Extension

IF I WAS WRITING THIS PAPER TODAY, WHAT WOULD I DO DIFFERENTLY? WHERE WOULD I INNOVATE?

Complexity Measure

- Though the Flesch Kincaid grade score seems to work decently in English, the scope of this project could expand if we were to use a measure that could compare across languages better.
- Though the comparisons are within countries, a lot of the data is filtered to be English, so it could add theoretical potential as well as more robust results.

- There might not be such a scale out there, so it might be worth theorizing how and if that could even be done.

New Data

- Since the release of this paper, both ParlSpeech and EUSpeech have had new versions drop. It would be worth using these newer versions to see if the relationships still hold.

- In addition to the new versions of the same data, I think it would be more interesting to see if these relationships hold across non WEIRD countries. We would need to find a reliable measure for language specific complexity, but if we are looking at within-country effects, that shouldn't be a problem.

Same Data – New Questions

- Are there specific shocks to countries that change the complexity of their speech?
  - Financial Catastrophes?
  - Terrorist Attacks?
- It would be interesting to see if personalities match on to speech complexity. In the theory section of the paper, they do mention the big five and the correlations with ideology.
  - Might need to consider which datasets this would be possible with. I saw some politician personality datasets, but I'm assuming less famous politicians might not be on there.
    - All that to say, maybe EUSpeech is the correct dataset for this.

Thank you!